

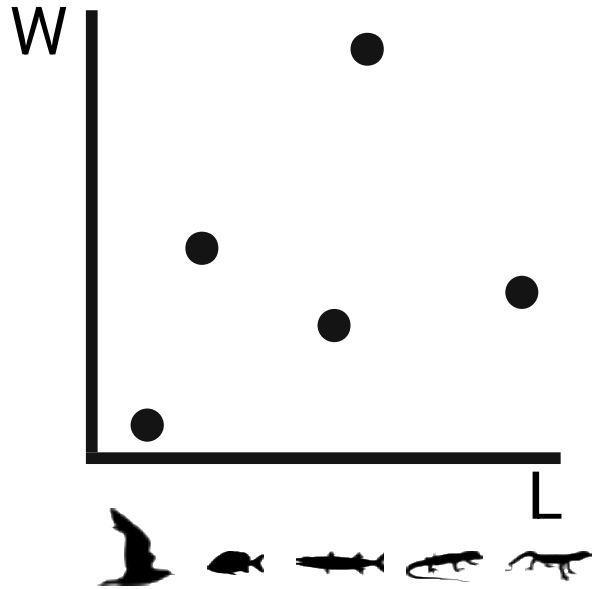
The non-parametric phylogenetic regressions via kernels

Ulises Rosas

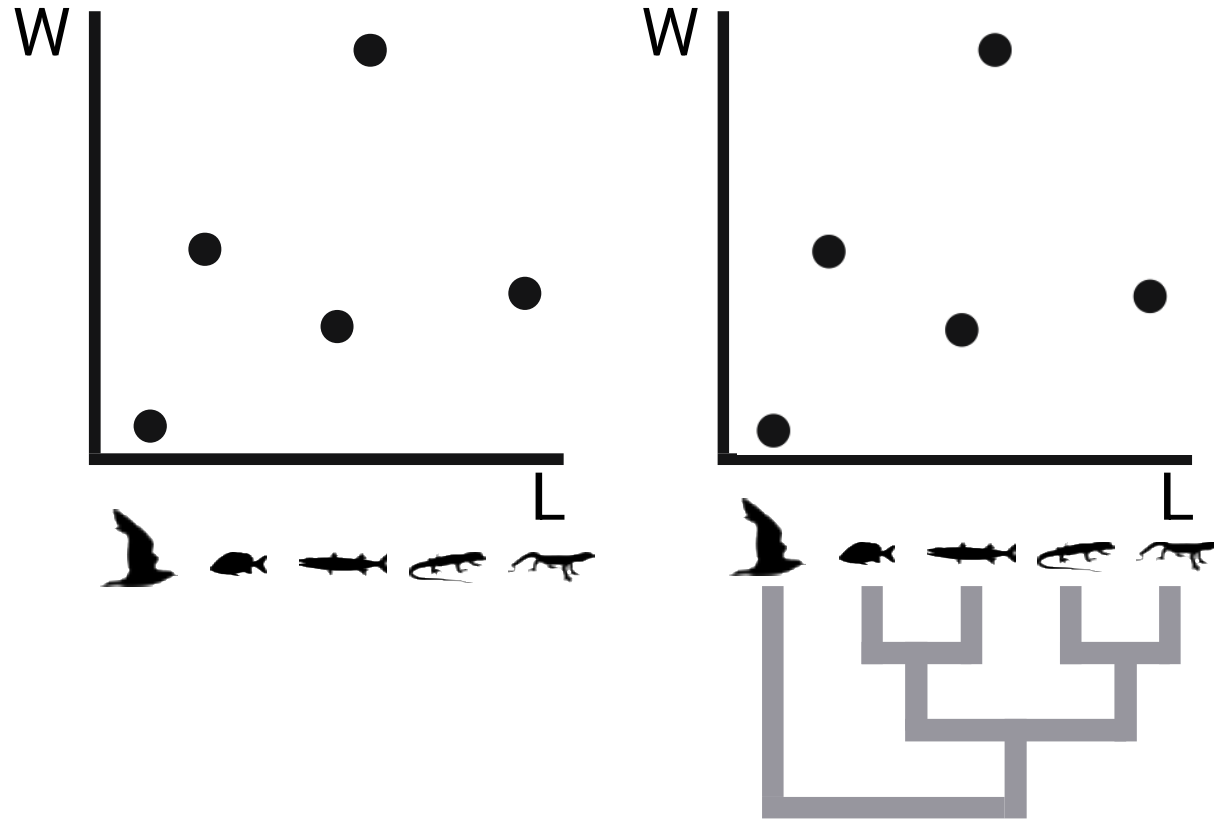
1. Background

- Phylogenetic Regression is a Generalized Least Squares (GLS)

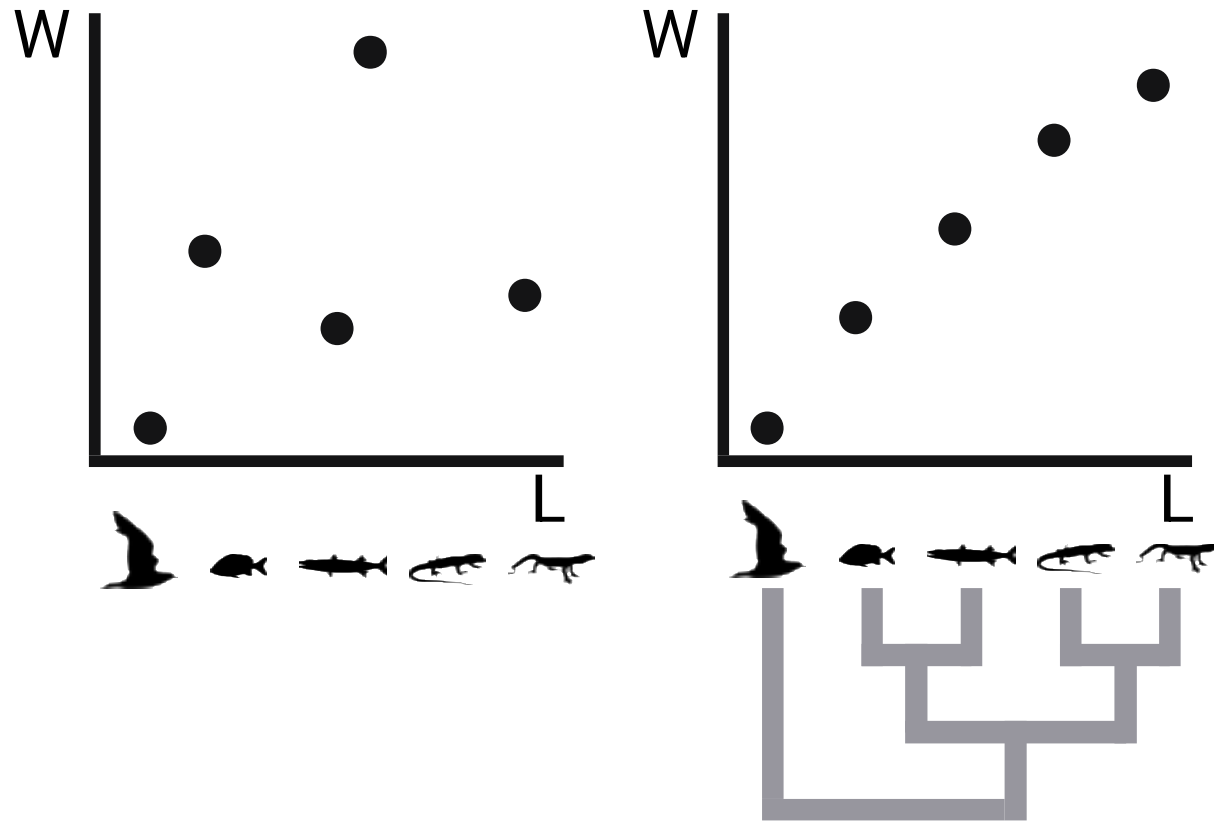
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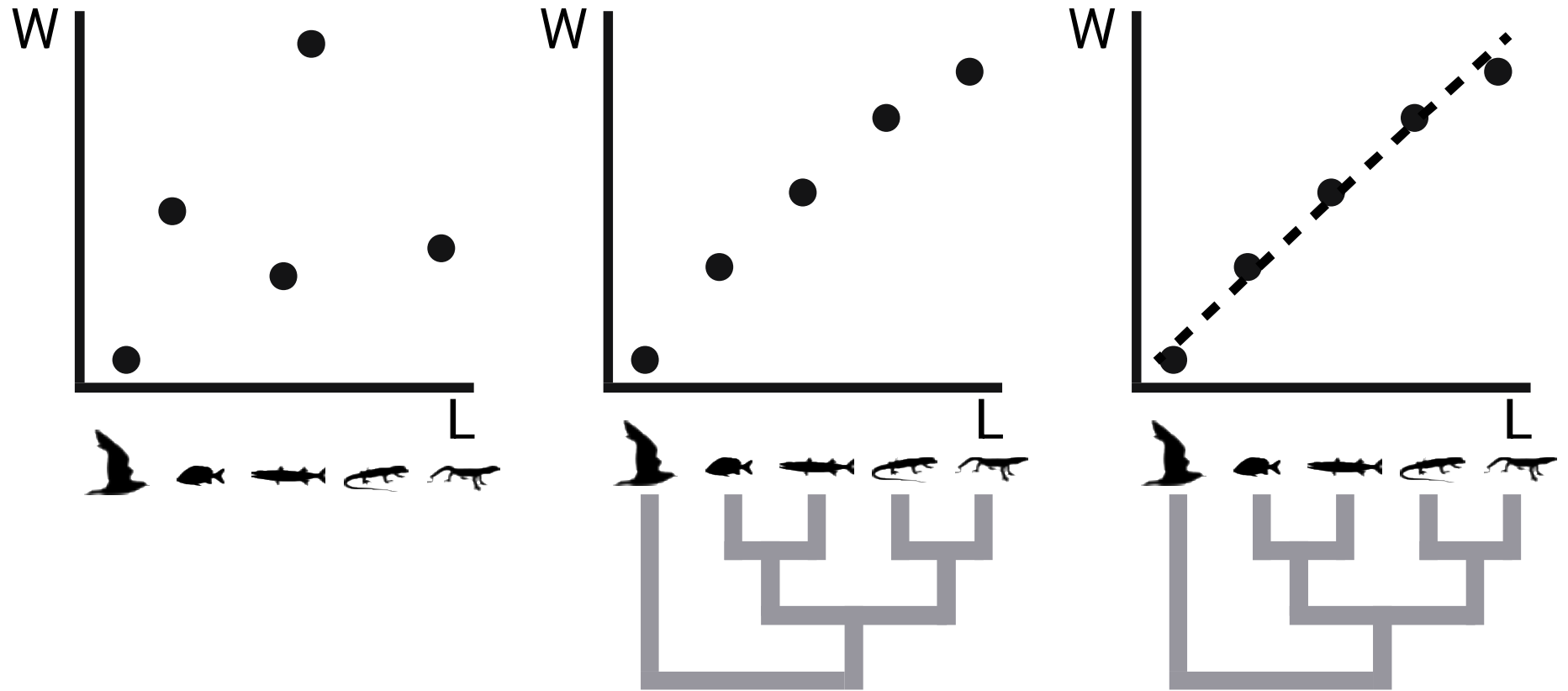
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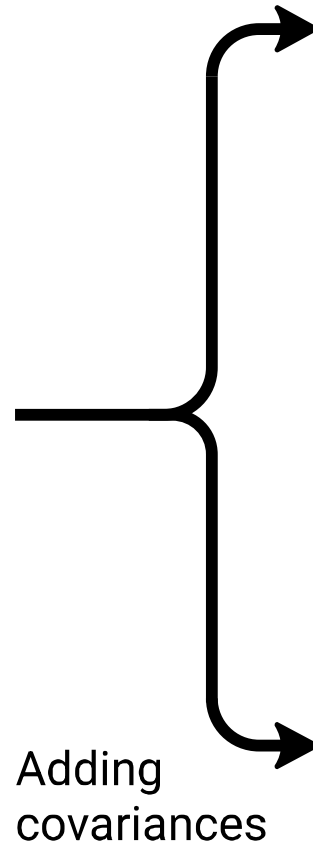
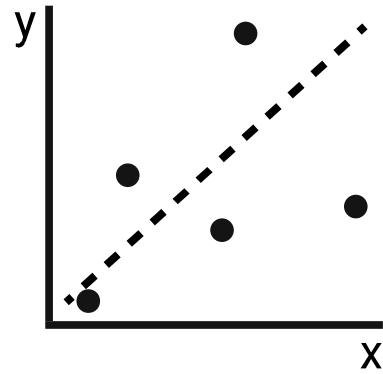
- Phylogenetic Regression is a Generalized Least Squares. The objective function $J(\beta)$ can be defined as:

$$\arg \min_{\beta} (\mathbf{Y} - \mathbf{X}\beta)^{\top} \mathbf{\Omega}^{-1} (\mathbf{Y} - \mathbf{X}\beta) = J(\beta)$$

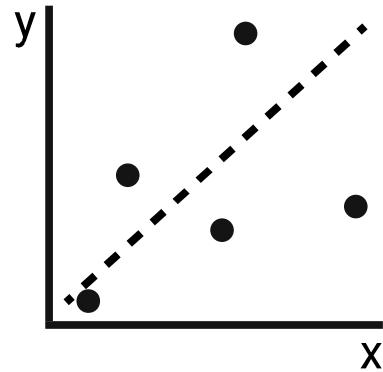
Diagram illustrating the components of the objective function $J(\beta)$:

- Response**: Points to $\mathbf{Y}$ (the response vector).
- Predictors**: Points to $\mathbf{X}$ (the predictor matrix).
- Phylogeny**: Points to $\mathbf{\Omega}^{-1}$ (the inverse of the covariance matrix).
- Unknown**: Points to β (the parameter vector to be estimated).

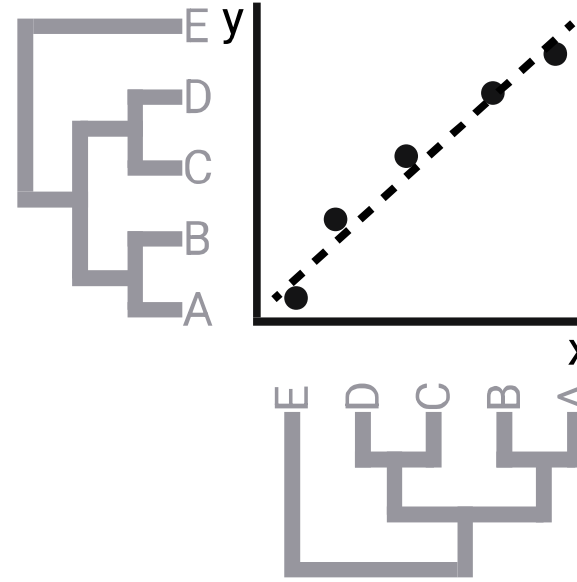
- However, data might not follow a linear relationship



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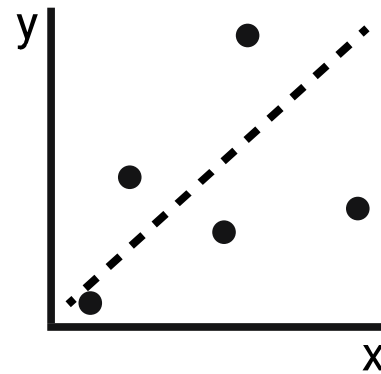


Adding
covariances

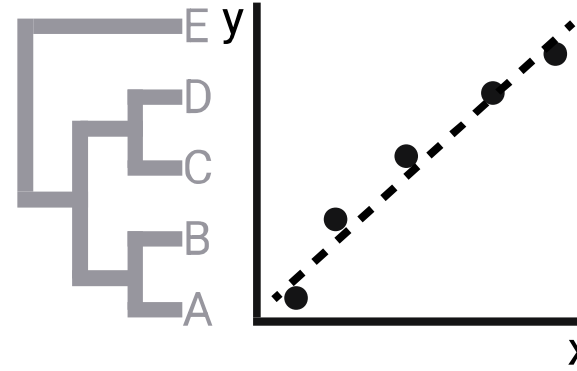


Linear
relationship

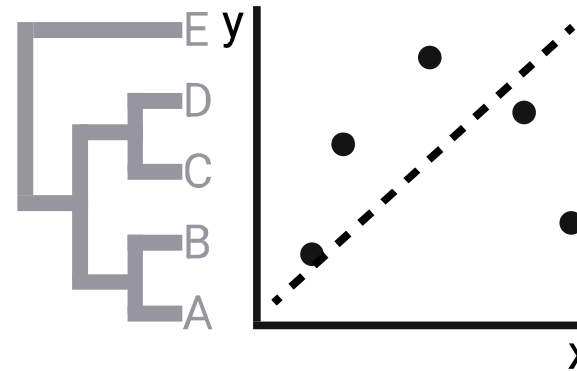
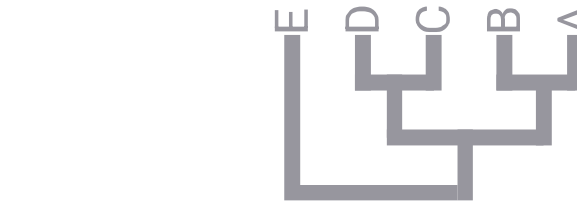
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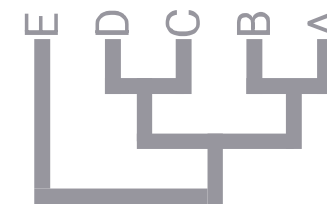
Adding
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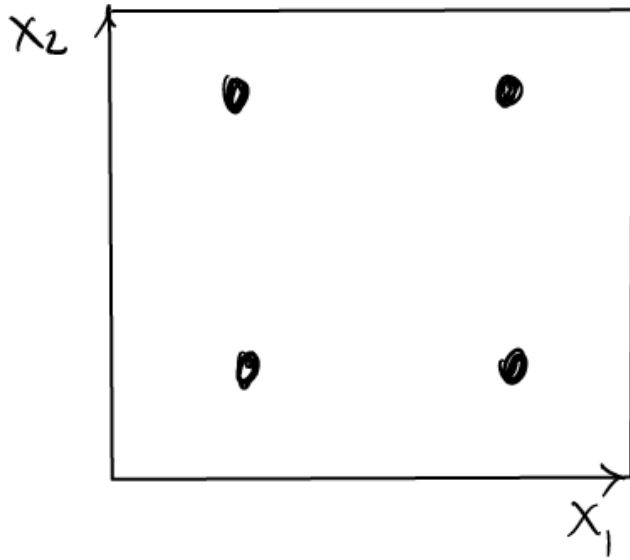
Linear
relationship



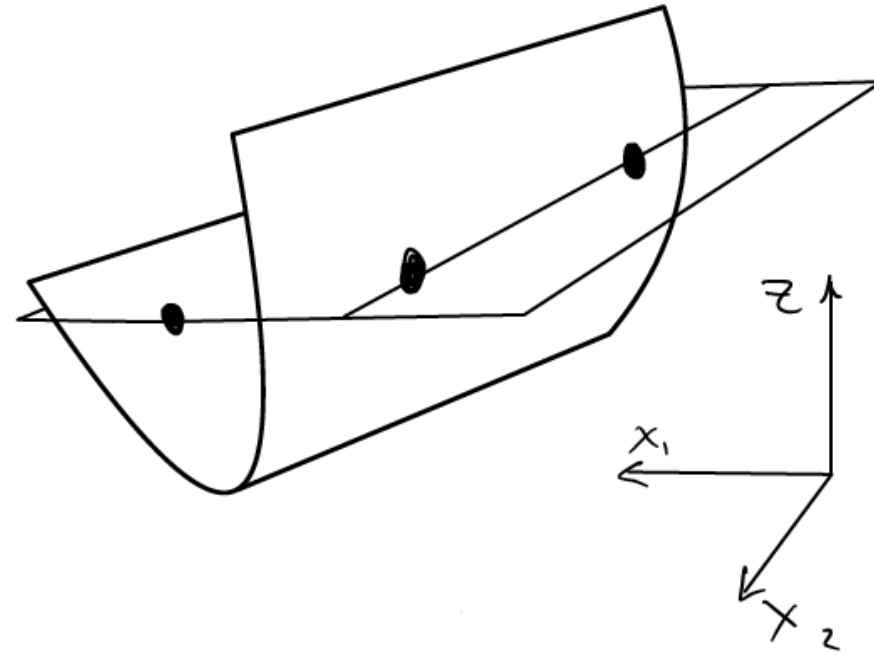
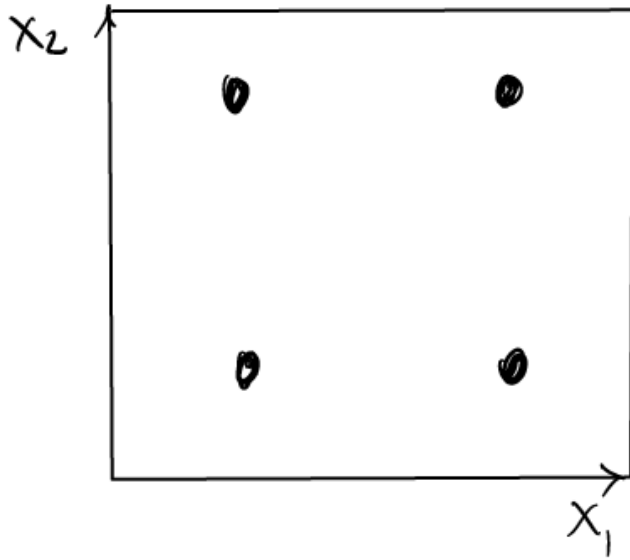
Non-Linear
relationship



- Then, I proposed the use of Regularized Kernel Regression to approach non-linearity in Phylogenetic Regressions



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Non-linear phylogenetic regression using regularised kernels

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Claudia Solís-Lemus⁴  | Ricardo Betancur-R^{1,5}

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⁴Wisconsin Institute for Discovery, Department of Plant Pathology, University of Wisconsin-Madison, Madison, Wisconsin, USA

⁵Scripps Institution of Oceanography, University of California San Diego, La Jolla, California, USA

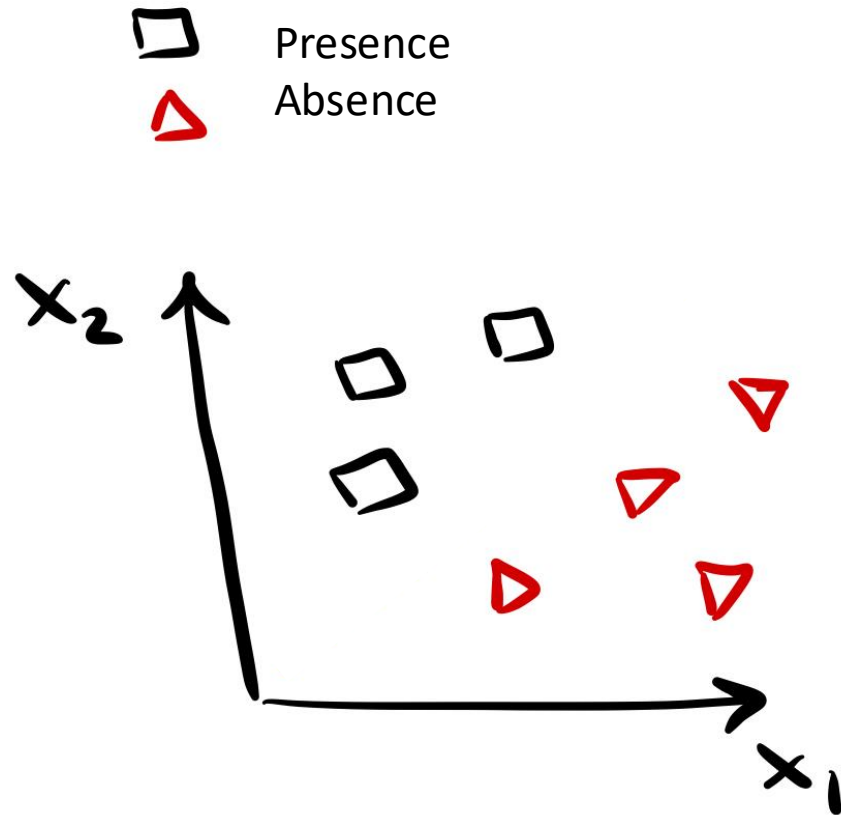
Abstract

1. Phylogenetic regression is a type of generalised least squares (GLS) method that incorporates a modelled covariance matrix based on the evolutionary relationships between species (i.e. phylogenetic relationships). While this method has found widespread use in hypothesis testing via phylogenetic comparative methods, such as phylogenetic ANOVA, its ability to account for non-linear relationships has received little attention.
2. To address this, here we implement a phylogenetic Kernel Ridge Regression (phyloKRR) method that utilises GLS in a high-dimensional feature space, employing linear combinations of phylogenetically weighted data to account for non-linearity. We analysed two biological datasets using the Radial Basis Function and linear kernel function.

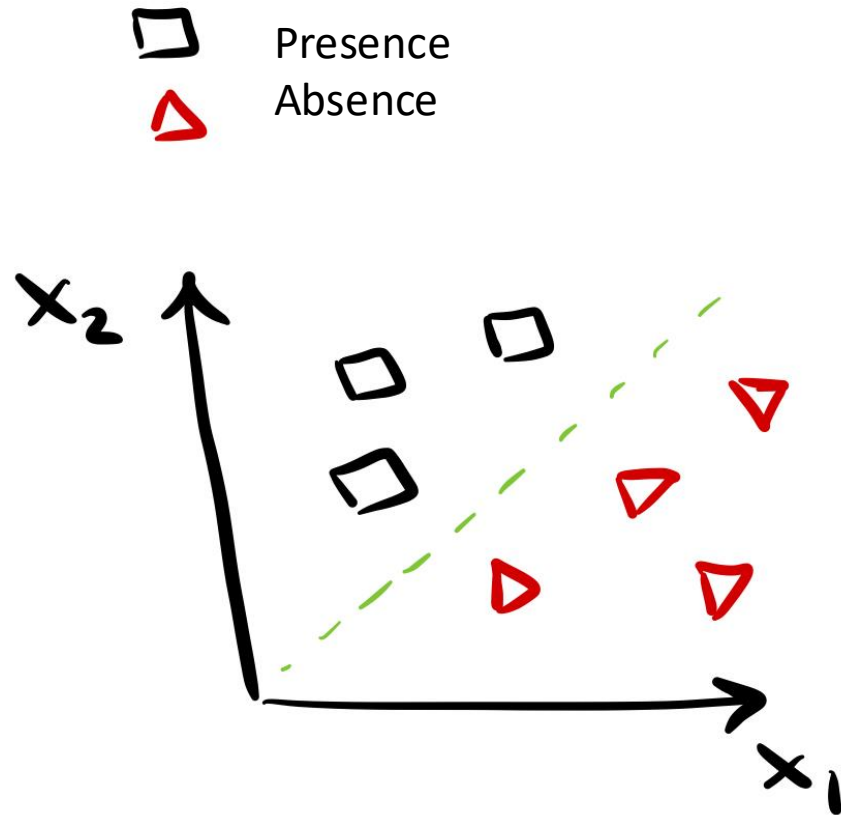
- **However**, it only works for continuous (*Gaussian*) responses

2. Generalizing the base model

What about other distributions?

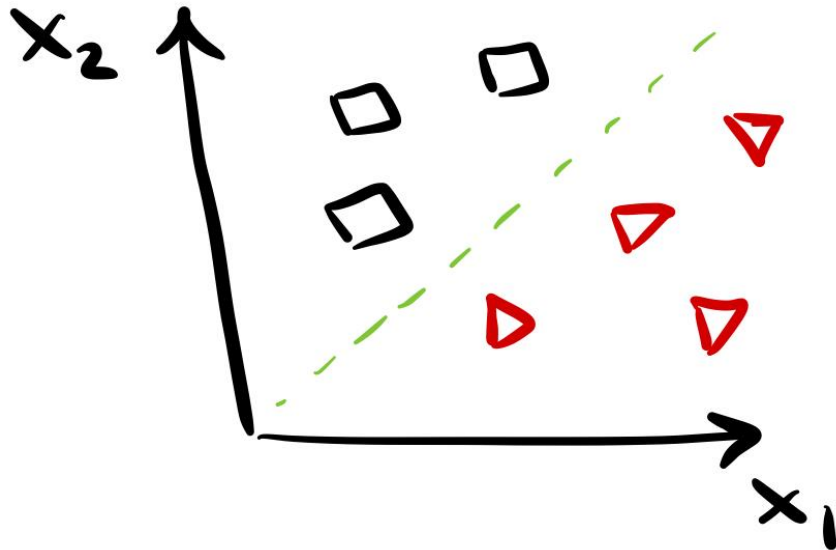


What about other distributions?

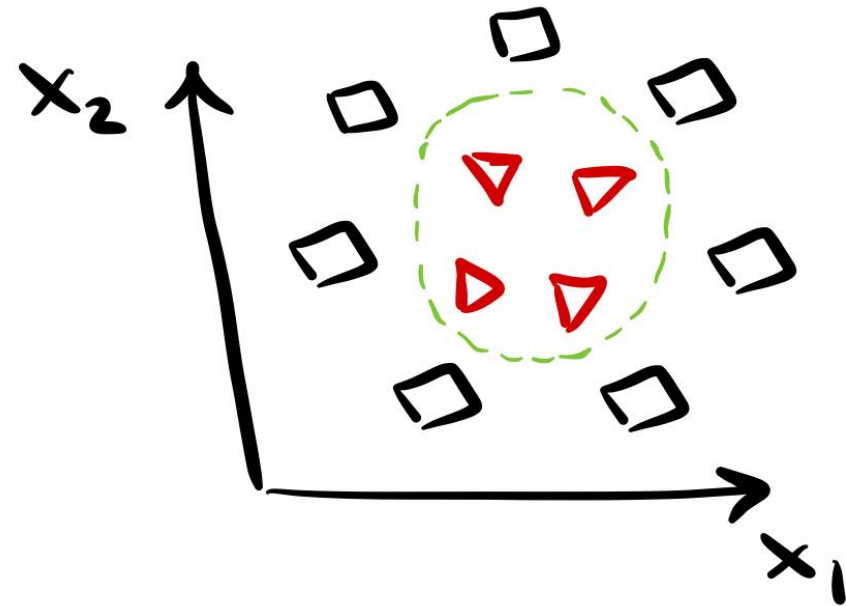


What about other distributions?

□ Presence
△ Absence



Linear relationship



Non-linear relationship

What about other distributions?

- **Binomial**: Presence/absence of traits (e.g., eye loss in fish)
 - **Poisson**: Counts, rates (e.g., number of fins, substitution rates)
 - **Gamma**: Proportions (e.g., male-female ratio)
 - Among others
-
- We need to use **GEE (Generalized Estimating Equations)**, a generalization of GLS.

Log-likelihood

vs

Quasi log-likelihood

$$l(y, \theta) = \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)$$

$$Q(y, \mu) = \int_y^{\mu} \frac{y - t}{\phi v(t)} dt$$

Log-likelihood

vs

Quasi log-likelihood

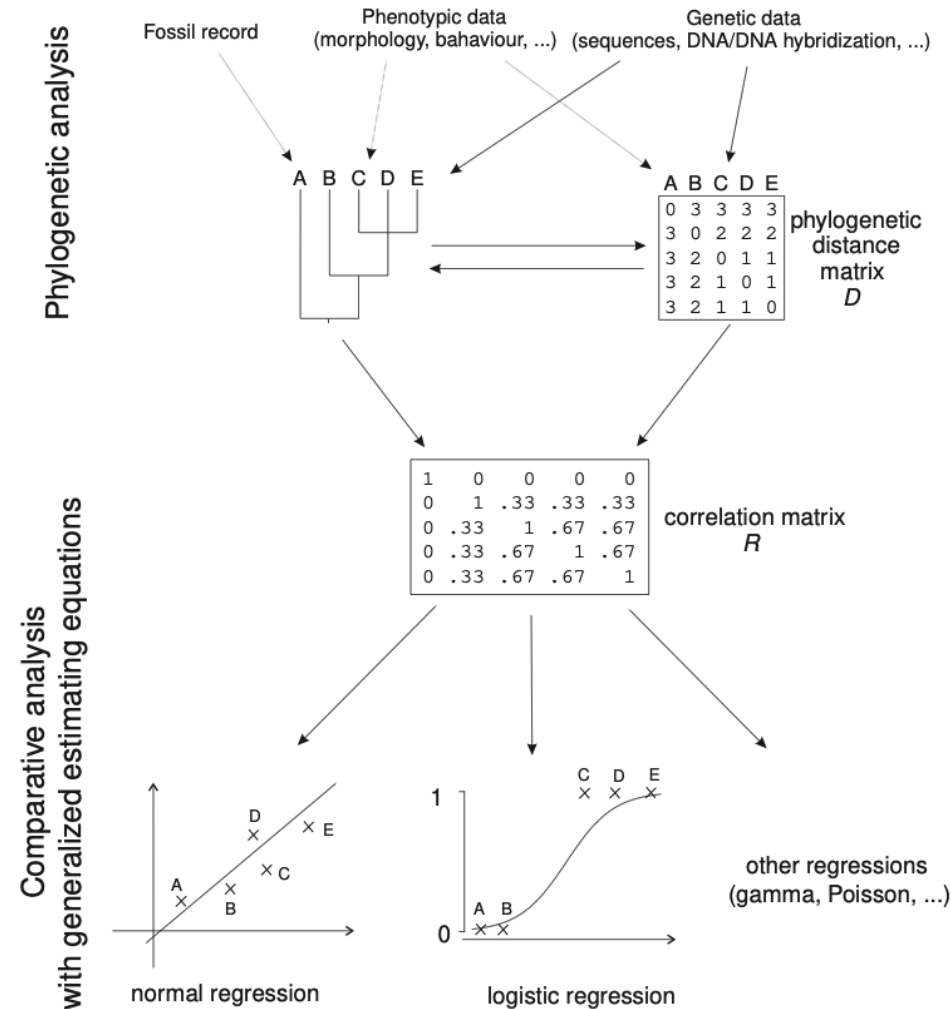
$$l(y, \theta) = \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)$$

- Score: $\frac{\partial l}{\partial \theta} = \frac{y - b'(\theta)}{a(\phi)}$
- All the exponential family assumptions
- We can **sometimes** include the phylogeny

$$Q(y, \mu) = \int_y^\mu \frac{y - t}{\phi v(t)} dt$$

- Score: $\frac{\partial Q}{\partial \mu} = \frac{y - \mu}{\phi v(\mu)}$
- Unique assumptions:
 - $E[y] = \mu$
 - $Var[y] = \phi v(\mu)$
- We can **always** include the phylogeny (GEE result)

What is Generalized Estimating Equations (GEE)?



New loss function from GEE

$$\underset{\boldsymbol{\beta} \in \mathbb{R}^p, \beta_0 \in \mathbb{R}}{\text{minimize}} \quad - \sum_{c=1}^n \int_{y_c}^{\mu_c} \frac{y_c - t}{\phi v(t)} dt + \frac{\lambda}{2} \|\boldsymbol{\beta}\|_2^2, \quad \mu_c = \sigma(\beta_0 + \mathbf{x}_c^\top \boldsymbol{\beta})$$

New loss function from GEE

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Kernelization

$$\underset{f \in \mathcal{H}_k, b \in \mathbb{R}}{\text{minimize}} \quad - \sum_{c=1}^n \int_{y_c}^{\mu_c} \frac{y_c - t}{\phi v(t)} dt + \frac{\lambda}{2} \|f\|_{\mathcal{H}_k}^2, \quad \mu_c = \sigma(\tilde{f}(\mathbf{x}_c))$$

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Representer Theorem

$$\tilde{f}(\mathbf{x}_c) = \sum_{i=1}^n \alpha_i k(\mathbf{x}_i, \mathbf{x}_c) + b$$

New loss function from GEE

$$\underset{\boldsymbol{\beta} \in \mathbb{R}^p, \beta_0 \in \mathbb{R}}{\text{minimize}} \quad - \sum_{c=1}^n \int_{y_c}^{\mu_c} \frac{y_c - t}{\phi v(t)} dt + \frac{\lambda}{2} \|\boldsymbol{\beta}\|_2^2, \quad \mu_c = \sigma(\beta_0 + \mathbf{x}_c^\top \boldsymbol{\beta})$$



Kernelization

$$\underset{f \in \mathcal{H}_k, b \in \mathbb{R}}{\text{minimize}} \quad - \sum_{c=1}^n \int_{y_c}^{\mu_c} \frac{y_c - t}{\phi v(t)} dt + \frac{\lambda}{2} \|f\|_{\mathcal{H}_k}^2 \quad \mu_c = \sigma(\tilde{f}(\mathbf{x}_c))$$



Representer Theorem

Non-parametric solution

New loss function from GEE

- In summary, the representer theorem solution:

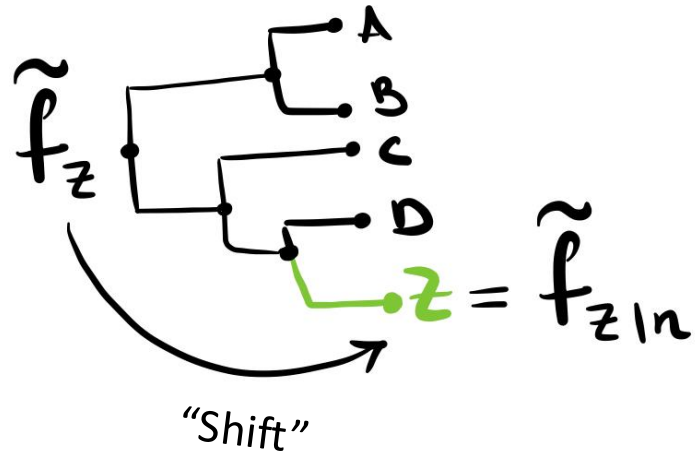
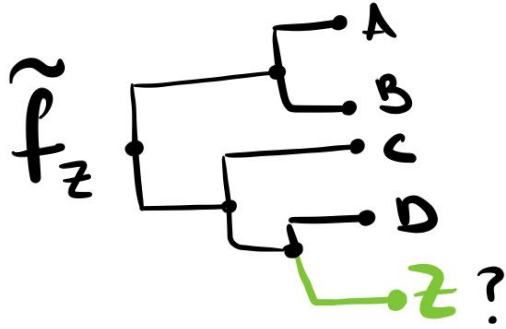
$$\tilde{f}(\mathbf{x}_c) = \sum_{i=1}^n \alpha_i k(\mathbf{x}_i, \mathbf{x}_c) + b$$

- Makes tractable the PQL in high-dimensional space:

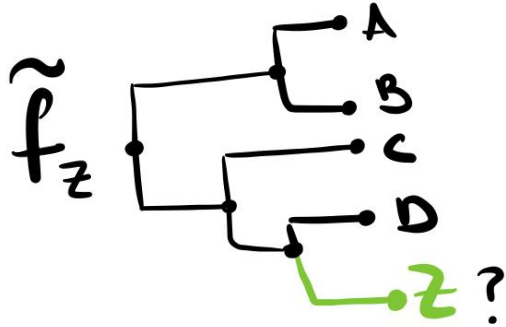
$$\underset{f \in \mathcal{H}_k, b \in \mathbb{R}}{\text{minimize}} \quad - \sum_{c=1}^n \int_{y_c}^{\mu_c} \frac{y_c - t}{\phi v(t)} dt + \frac{\lambda}{2} \|f\|_{\mathcal{H}_k}^2$$

- Remarkably, we can get derivatives and, thus, minimize the new loss function.

Adjusted predictions for the model



Adjusted predictions for the model



$$\begin{bmatrix} \sigma^{-1}(\mathbf{y}_n) \\ \sigma^{-1}(\mathbf{y}_z) \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \tilde{\mathbf{f}}_n \\ \tilde{\mathbf{f}}_z \end{bmatrix}, \begin{bmatrix} \mathbf{R}_{nn} & \mathbf{R}_{zn}^\top \\ \mathbf{R}_{zn} & \mathbf{R}_{zz} \end{bmatrix} \right)$$



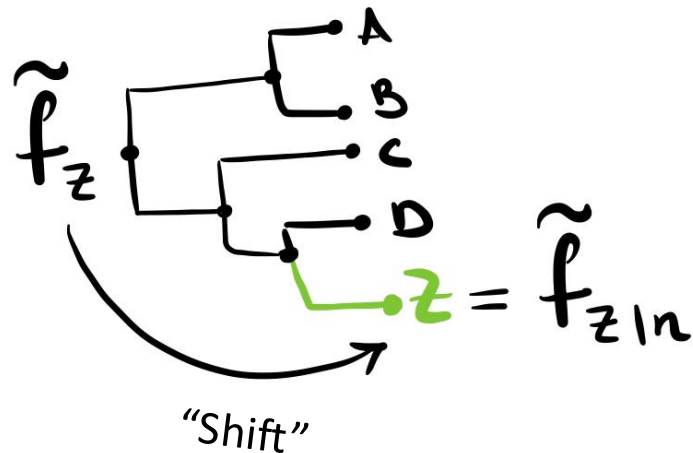
Conditional mean theorem

$$\tilde{\mathbf{f}}_{z|n} = \tilde{\mathbf{f}}_z + \mathbf{R}_{zn} \mathbf{R}_{nn}^{-1} \left\{ \sigma^{-1}(\mathbf{y}_n) - \tilde{\mathbf{f}}_n \right\}$$

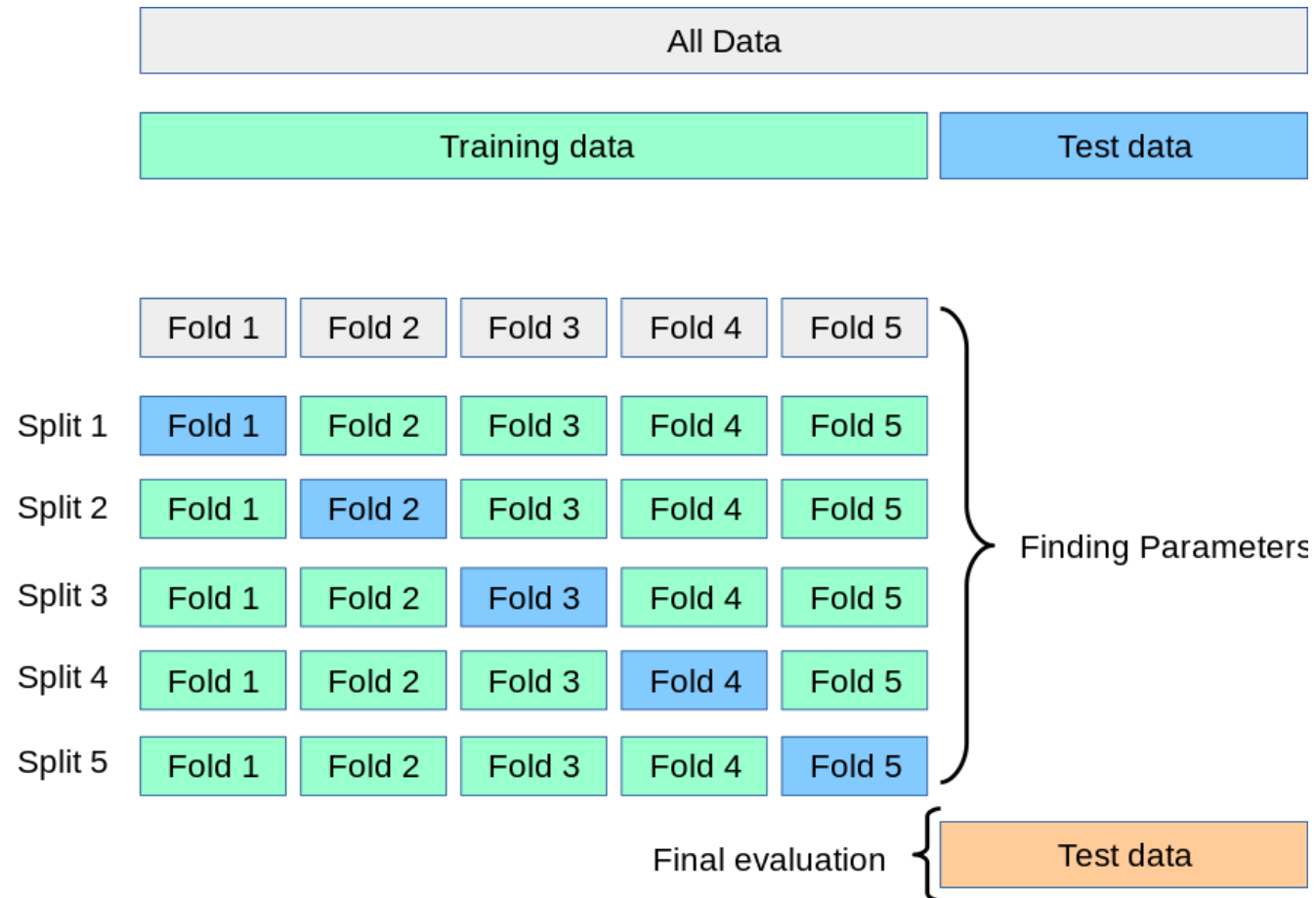


Plugging it into the link function

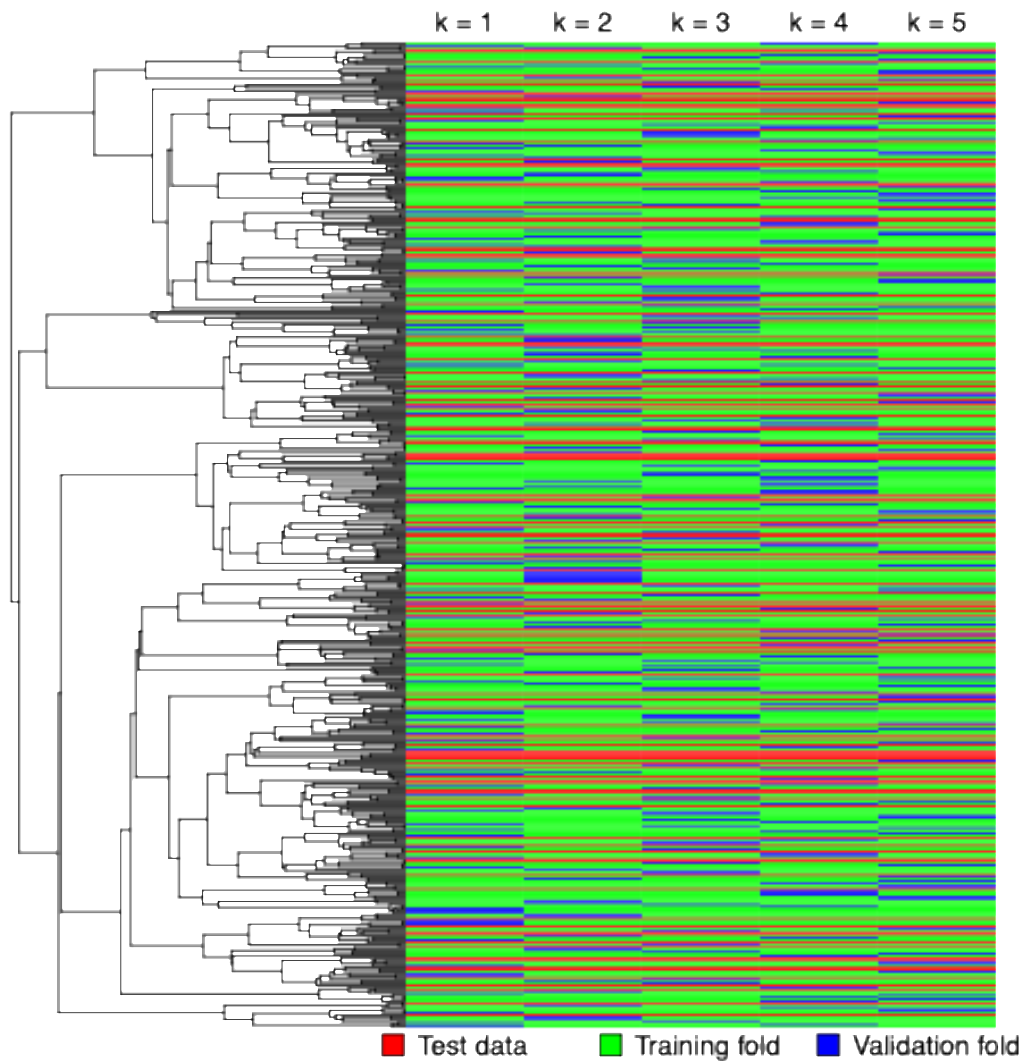
$$\mu_{z|n} = \sigma \left(\tilde{\mathbf{f}}_{z|n} \right)$$



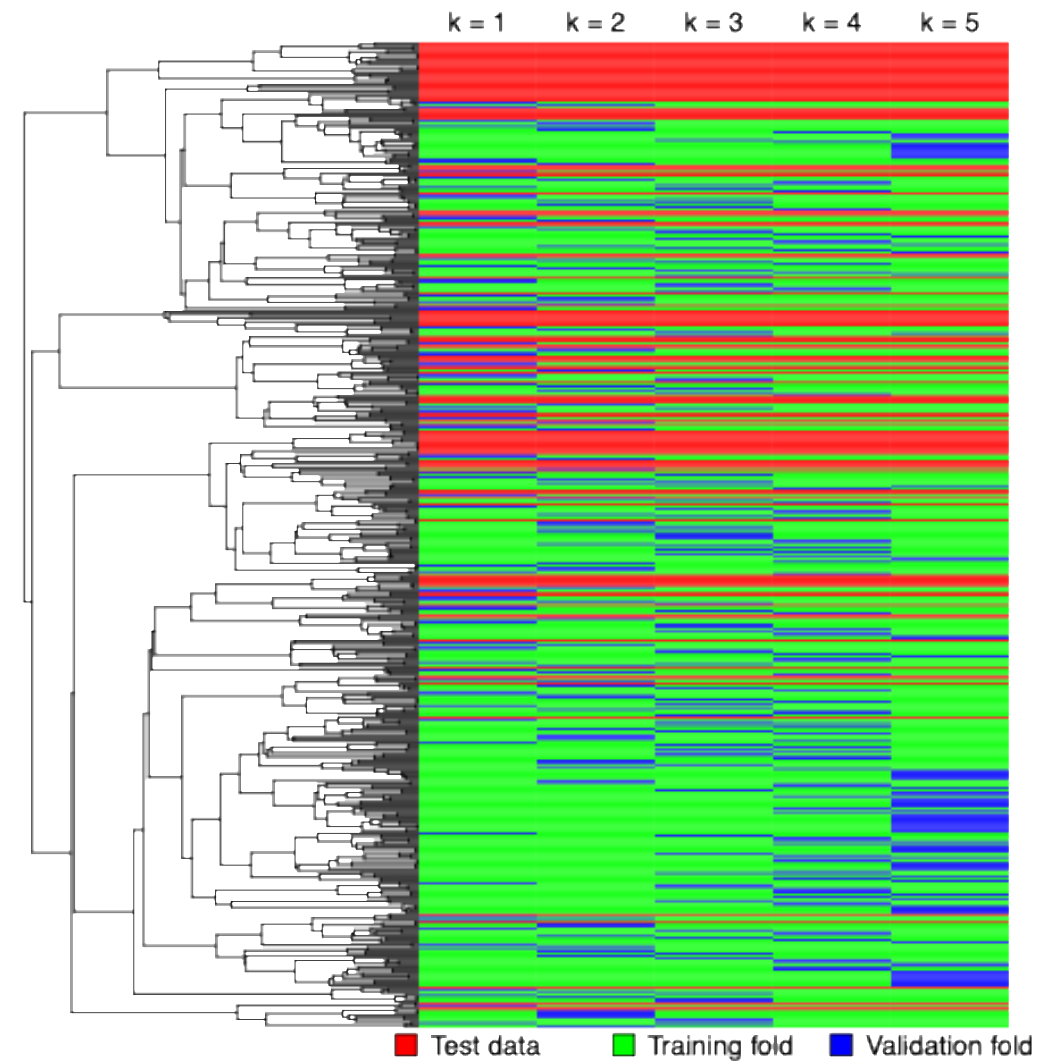
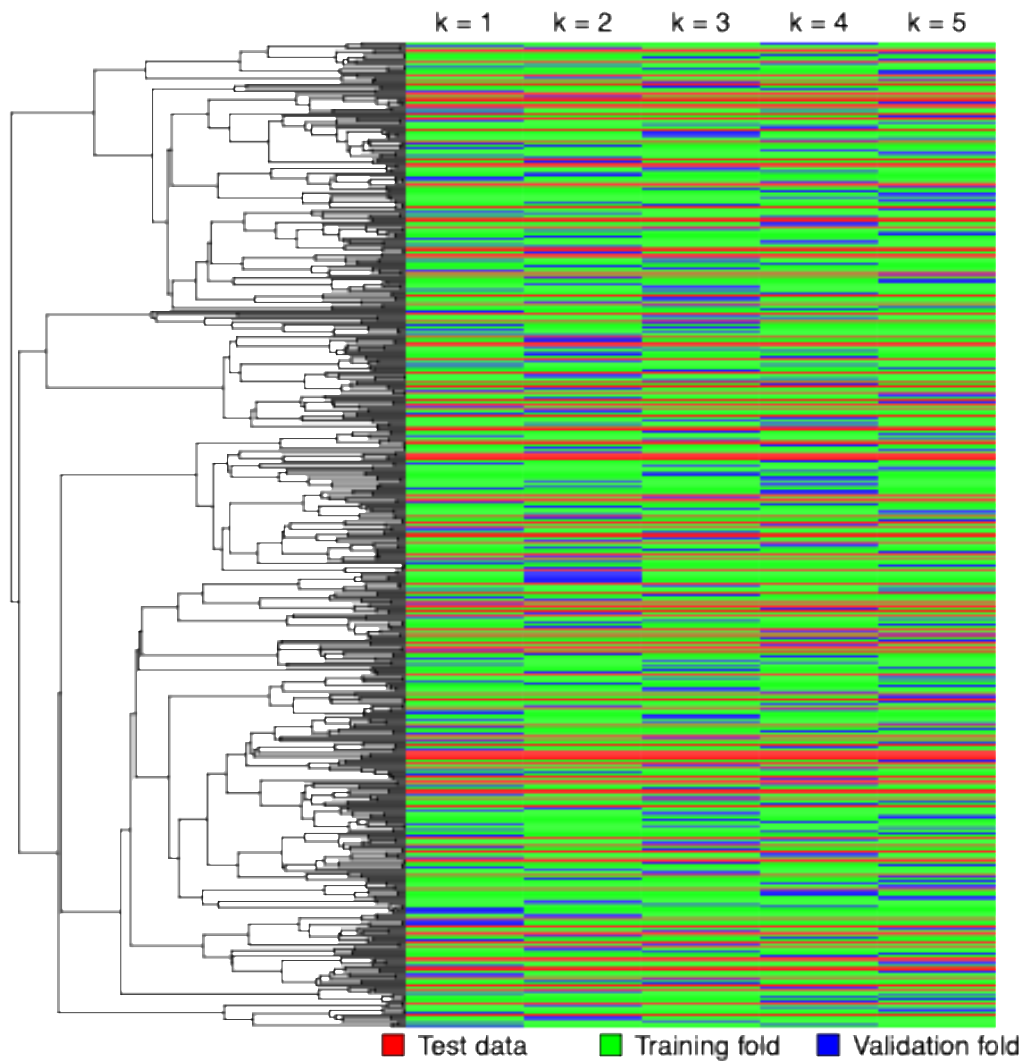
Modified cross-validation



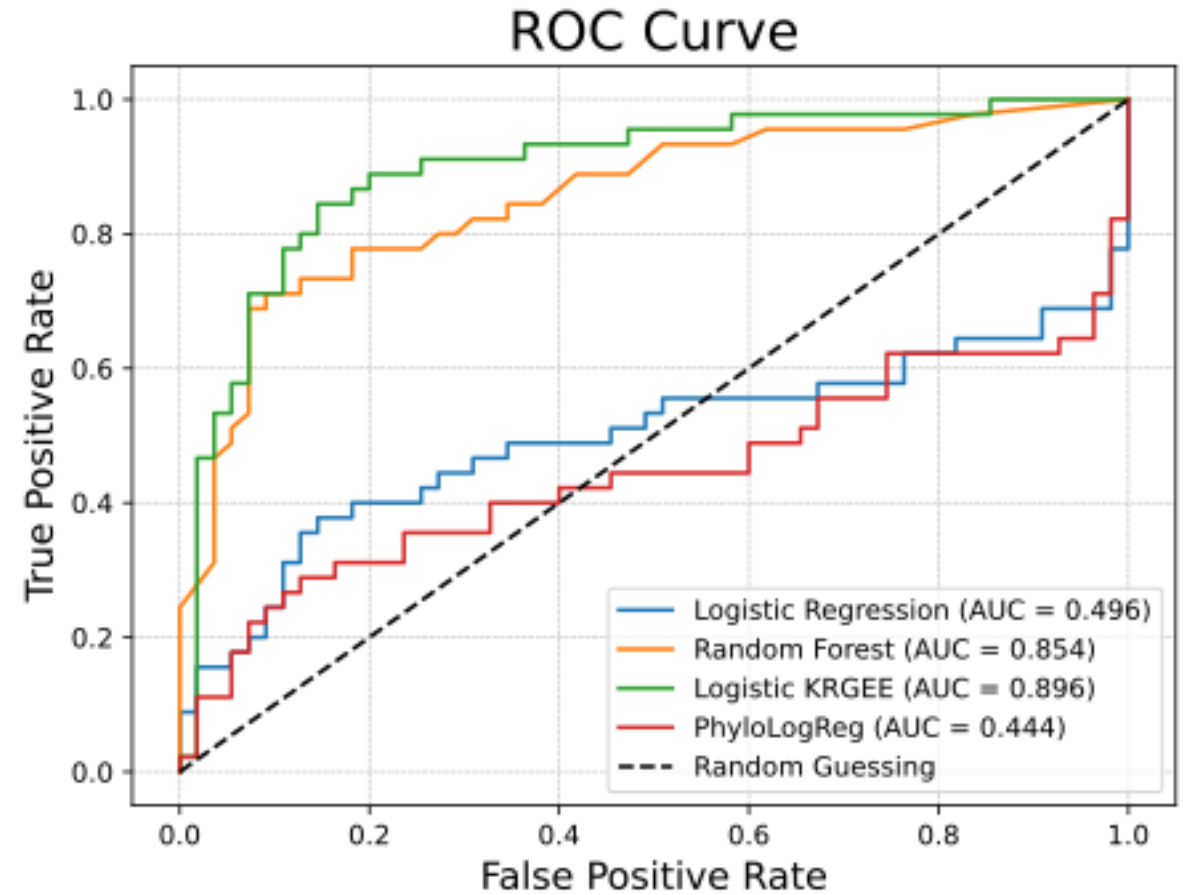
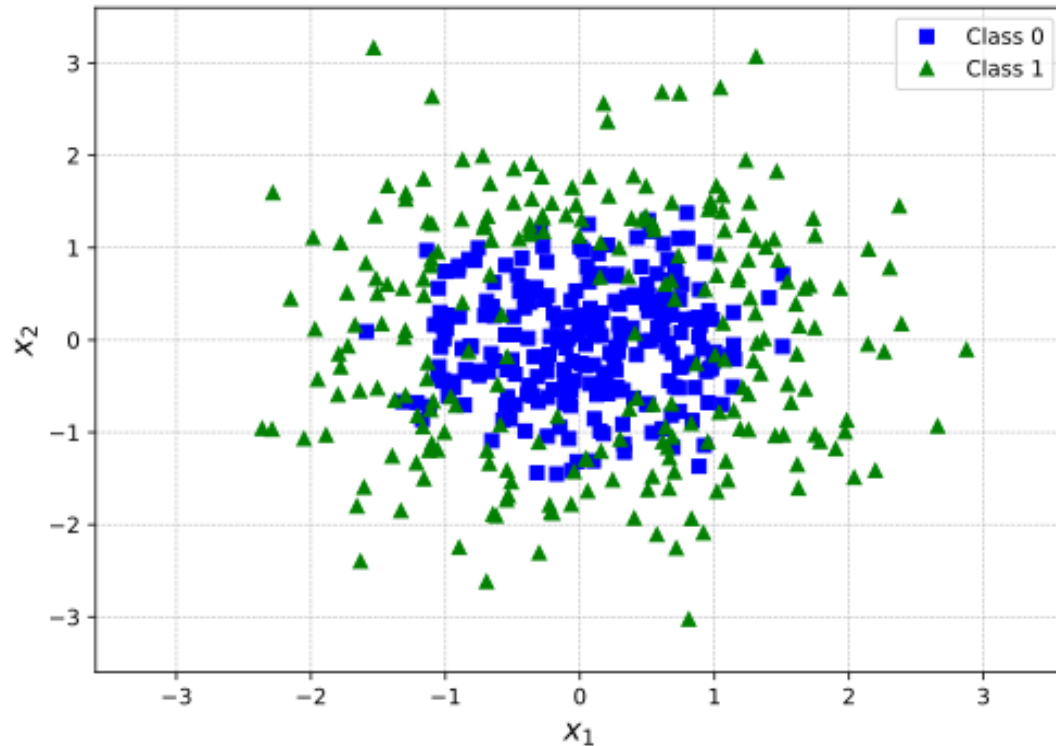
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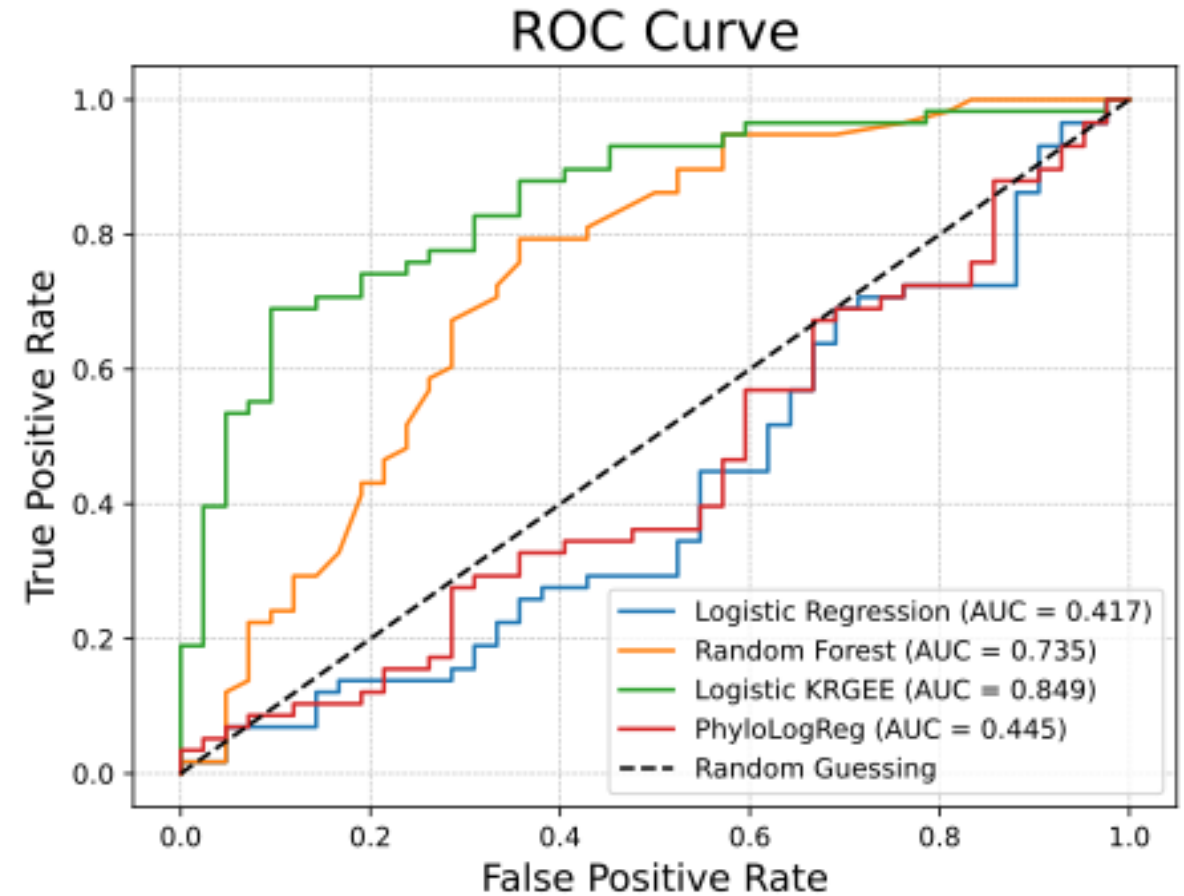
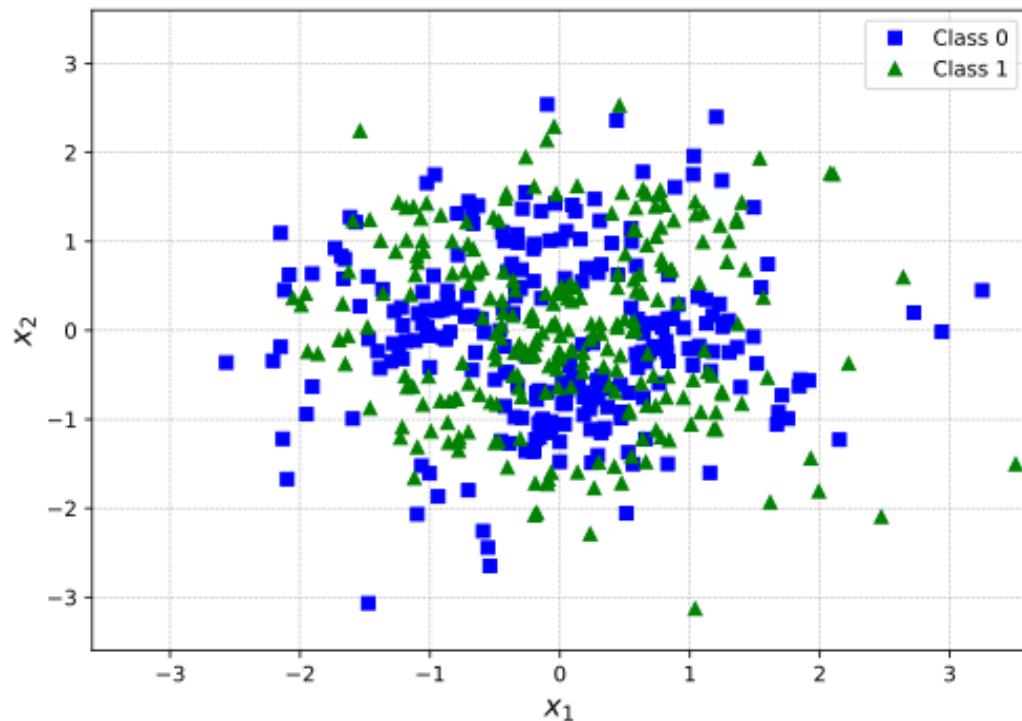
Modified cross-validation



Simulated dataset with correlated observations



Simulated dataset with correlated observations



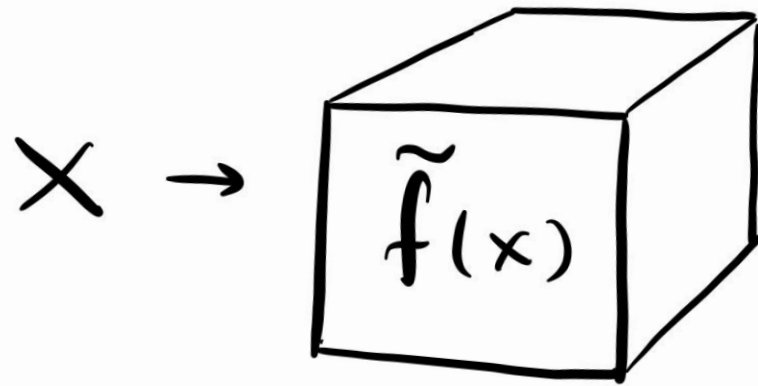
So far...

- We have created a fully non-parametric phylogenetic regression.
- It is statistically principled using kernels.
- **However**, it is a black box.



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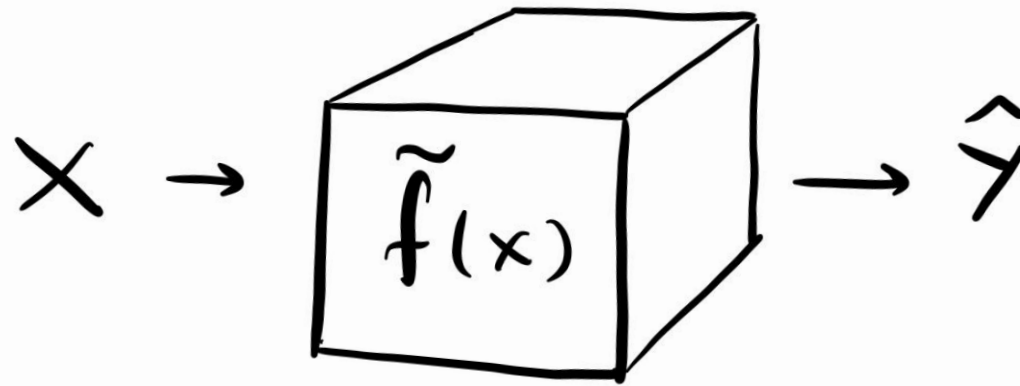
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*Something
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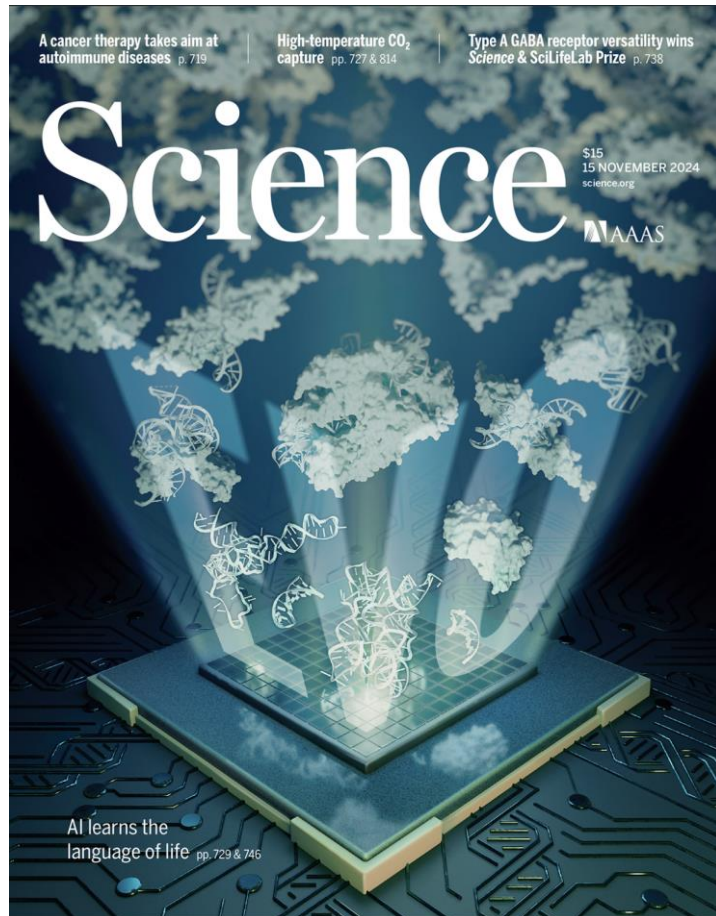
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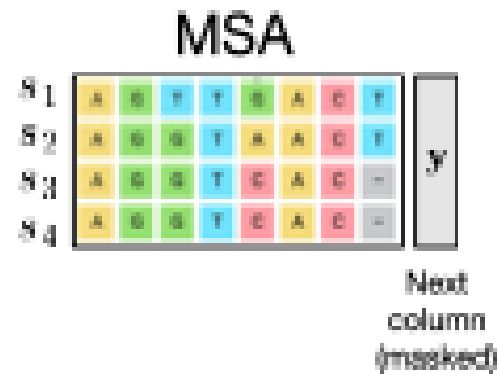
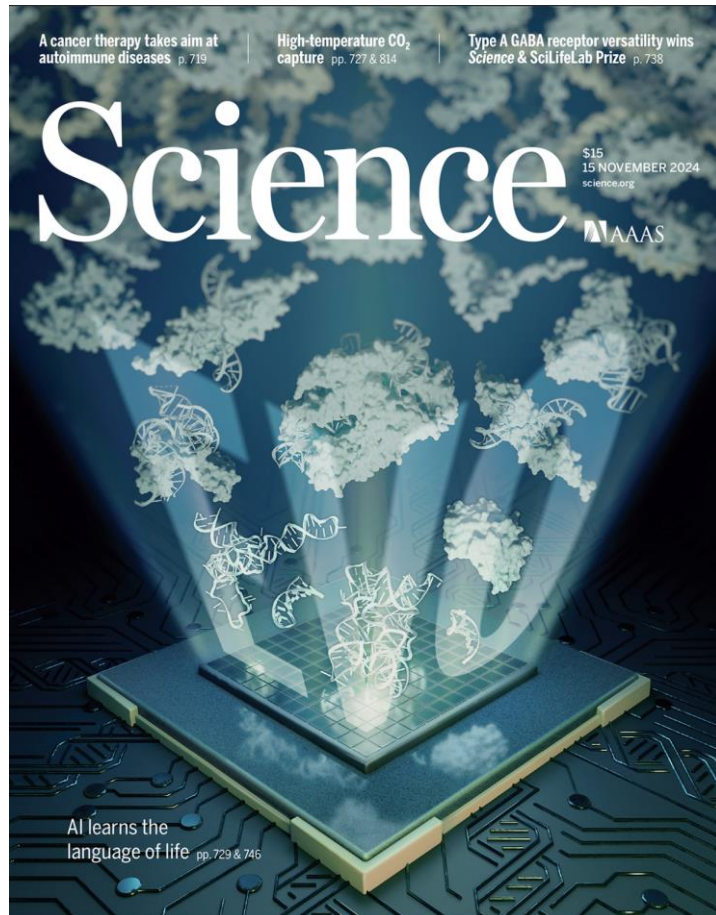


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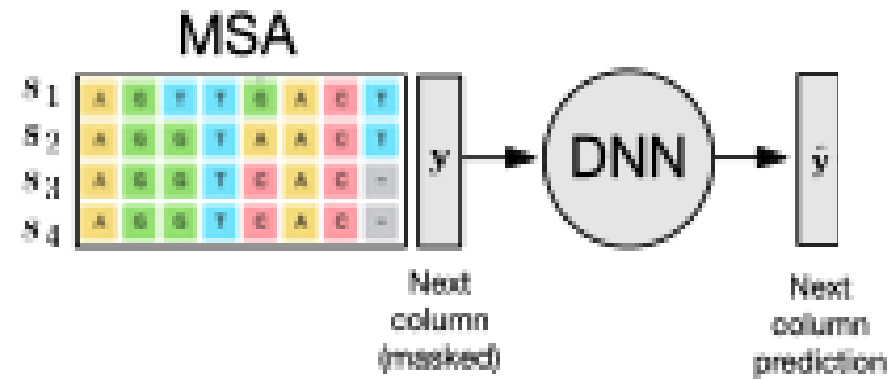
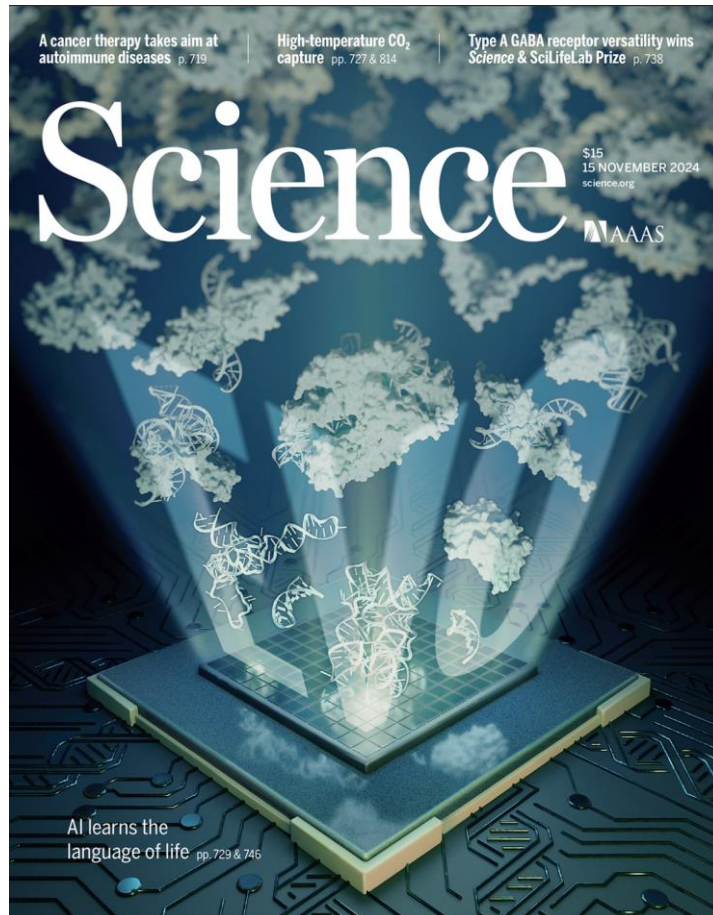
Genome Language Models



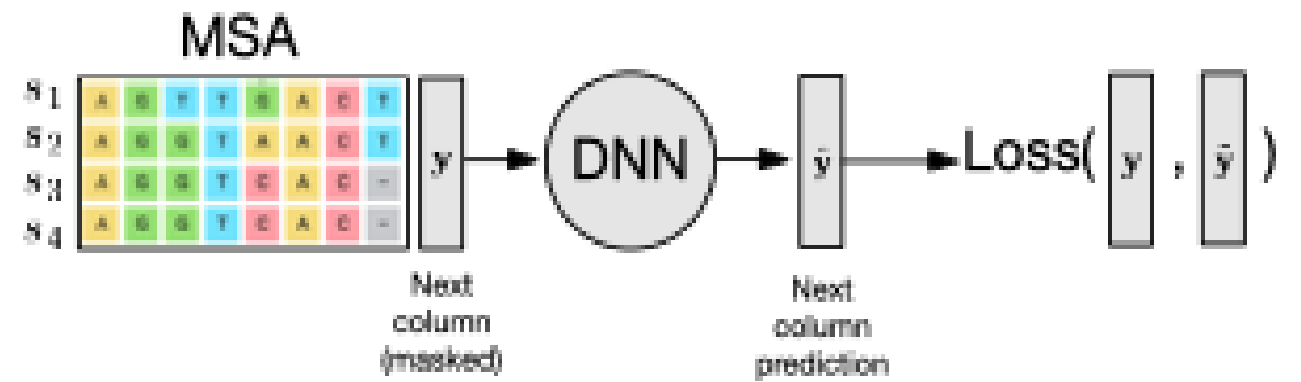
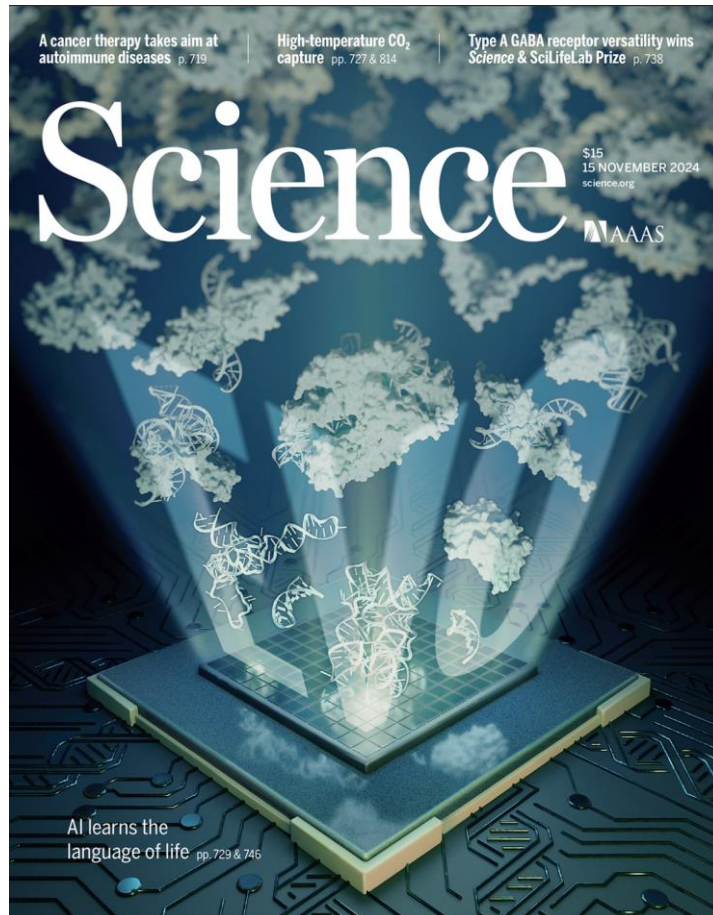
Genome Language Models



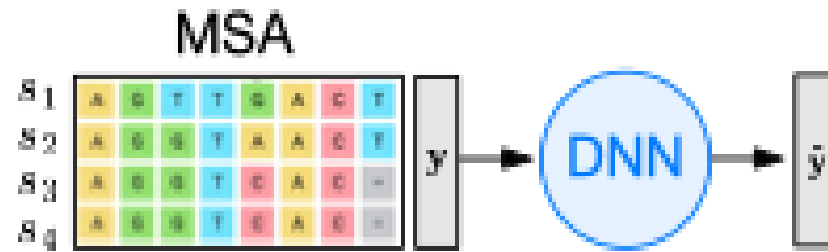
Genome Language Models



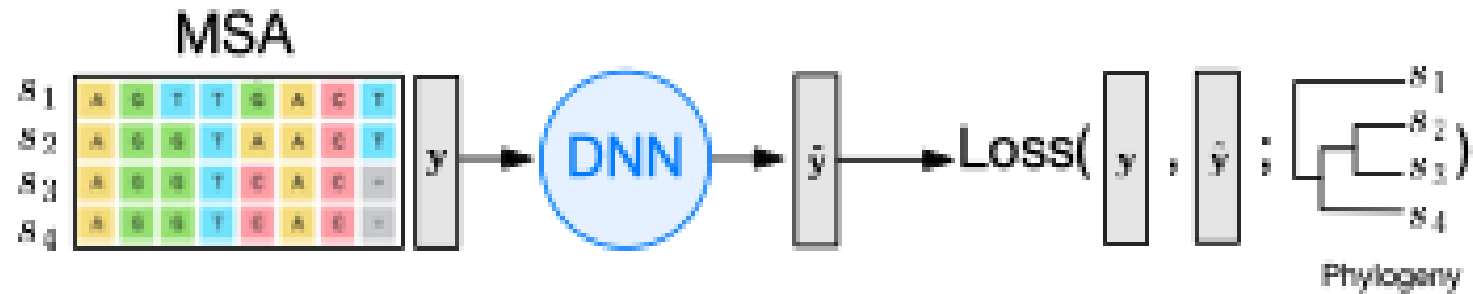
Genome Language Models



(Phylogenetic) Genome Language Models



(Phylogenetic) Genome Language Models



New foundational model

Conclusions

- The non-parametric phylogenetic regression accounts for complex relationships.
- On the development, it proposes a new loss function.
- This loss function can be repurposed for other non-parametric methods (e.g., deep learning).

Conclusions

- Assumes there exists 1 single correlation structure.
- Future work:
 - An ensemble of correlation structures, including the identity.
 - *Open question*: How to adjust predictions under other stochastic processes (e.g., OU, Brownian Bridge)?
- Increase efforts to connect standard comparative methods with traditional machine learning techniques.

- Do you have other possible applications in mind?

Thanks for your attention!

- Collaborators:
 - Siavash Mirarab
 - Matt Pennell

